

Design Verification Report

Short-Circuit Analysis & Protection Coordination

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1 Introduction

This report details the short-circuit analysis and protection coordination study for the Al-Nour Developments 8-story commercial building. As mandated, a manual benchmark verification of the worst-case critical path (Floor 8) is performed utilizing the IEC 60909 equivalent voltage source methodology.

2 Software Overview & Modeling Assumptions

To fulfill the detailed engineering requirements, Schneider Electric Ecodial was utilized as the primary power system analysis software. The software engine intrinsically adheres to IEC 60909 for short-circuit calculations and IEC 60364 for cable sizing and thermal constraints.

The digital simulation model of the Al-Nour Developments commercial network was constructed under the following fundamental assumptions and input parameters:

- **System Topology:** 400/230V, 3-Phase, 4-Wire distribution utilizing a strict TN-S earthing configuration.
- **Utility Source Parameters:** Modeled at the Main Distribution Board (MDB) incoming terminals with an updated maximum three-phase fault capacity of $I_{k3} = 20$ kA and a single-phase earth fault capacity of $I_{k1} = 15$ kA.
- **Load Estimations:** Each Sub-Main Distribution Board (SMDB) was allocated a lumped continuous load of 150 kVA to represent typical office lighting, receptacles, and localized HVAC machinery.
- **Routing & Impedance Assumptions:** Vertical and horizontal cable lengths were estimated based on typical 8-story commercial architecture, with the critical path targeted at the topmost Floor 8. Feeder cross-sectional areas were sized dynamically by the software to handle the 150 kVA floor loads while maintaining voltage drop compliance ($\Sigma\Delta U < 4\%$).

3 Benchmark Verification: Worst-Case Critical Path

The following calculations determine the initial symmetrical three-phase (I''_{k3}) and single-phase earth (I''_{k1}) fault currents at the furthest final Distribution Board on Floor 8. To establish the benchmark targets, the Ecodial software simulation output at the DB busbar (Node WC 31 / QA 51) is presented below in Figure 1.

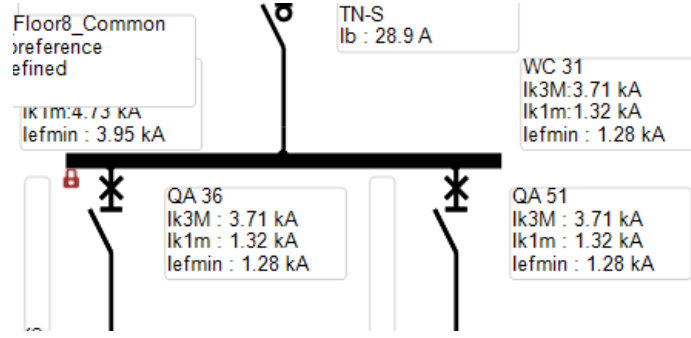


Figure 1: Ecodial simulation output at the Floor 8 Common DB busbar. The target values for manual verification are $I_{k3M} = 3.71$ kA and $I_{k1m} = 1.32$ kA.

3.1 Step 1: Equivalent Source Impedance (Z_{source})

To establish the mathematical baseline, the utility grid and incoming transformer are modeled as an equivalent static Thévenin voltage source at the incoming terminals of the Main Distribution Board (MDB).

Per the design parameters, the maximum incoming three-phase short-circuit current is $I_{k3} = 20$ kA, and the nominal system voltage is $U_n = 400$ V. For maximum fault calculations, the IEC 60909 voltage correction factor is $c = 1.05$.

The total magnitude of the source impedance is calculated as:

$$Z_{source} = \frac{c \cdot U_n}{\sqrt{3} \cdot I_{k3}} \quad (1)$$

$$Z_{source} = \frac{1.05 \cdot 400}{\sqrt{3} \cdot 20000} = \frac{420}{34641.02} \approx 0.01212 \Omega \quad (12.12 \text{ m}\Omega) \quad (2)$$

For high-capacity low-voltage utility feeds, the network is highly inductive. Assuming a standard X/R ratio of 10 (where $X = 10R$), we apply the impedance magnitude formula:

$$Z_{source} = \sqrt{R_{source}^2 + X_{source}^2} = \sqrt{R_{source}^2 + (10R_{source})^2} = R_{source} \sqrt{101} \quad (3)$$

Solving for the positive-sequence resistance (R_{source}) and reactance (X_{source}):

$$R_{source} = \frac{12.12}{\sqrt{101}} \approx 1.206 \text{ m}\Omega \quad (4)$$

$$X_{source} = 10 \cdot R_{source} \approx 12.06 \text{ m}\Omega \quad (5)$$

These values constitute the zero-node baseline at the MDB busbar. The cumulative impedances of the downstream internal network will be added sequentially to this baseline.

3.2 Step 2: Cumulative Cable Impedances

To determine the fault attenuation along the critical path, the positive-sequence resistance (R) and reactance (X) for each distribution cable segment must be calculated.

Per IEC 60909 methodology for maximum short-circuit currents, the resistivity of copper at a baseline temperature of 20°C ($\rho = 18.5 \text{ m}\Omega \cdot \text{mm}^2/\text{m}$) is utilized to capture the absolute lowest physical resistance. The standard reactance parameter for multi-core low-voltage cables ($x = 0.08 \text{ m}\Omega/\text{m}$) is applied.

The standard impedance equations are:

$$R = \frac{\rho \cdot L}{A} \quad (6)$$

$$X = x \cdot L \quad (7)$$

Segment 1: MDB to Floor 8 SMDB

Parameters: Length (L_1) = 43 m, Cross-Sectional Area (A_1) = 120 mm²

$$R_1 = \frac{18.5 \cdot 43}{120} \approx 6.63 \text{ m}\Omega \quad (8)$$

$$X_1 = 0.08 \cdot 43 = 3.44 \text{ m}\Omega \quad (9)$$

Segment 2: Floor 8 SMDB to Final DB

Parameters: Length (L_2) = 30 m, Cross-Sectional Area (A_2) = 10 mm²

$$R_2 = \frac{18.5 \cdot 30}{10} = 55.50 \text{ m}\Omega \quad (10)$$

$$X_2 = 0.08 \cdot 30 = 2.40 \text{ m}\Omega \quad (11)$$

3.3 Step 3: Initial Symmetrical Three-Phase Short-Circuit Current (I''_{k3})

To calculate the worst-case three-phase fault at the Floor 8 Final Distribution Board busbar, the cumulative positive-sequence impedance of the source and the two internal feeder segments (MDB to SMDB, and SMDB to DB) are summated.

The total network resistance (R_{total}) is:

$$R_{total} = R_{source} + R_1 + R_2 \quad (12)$$

$$R_{total} = 1.206 + 6.63 + 55.50 = 63.336 \text{ m}\Omega \approx 63.34 \text{ m}\Omega \quad (13)$$

The total network reactance (X_{total}) is:

$$X_{total} = X_{source} + X_1 + X_2 \quad (14)$$

$$X_{total} = 12.06 + 3.44 + 2.40 = 17.90 \text{ m}\Omega \quad (15)$$

Applying the AC impedance magnitude formula:

$$|Z_k| = \sqrt{R_{total}^2 + X_{total}^2} = \sqrt{63.34^2 + 17.90^2} = \sqrt{4011.96 + 320.41} \approx 65.82 \text{ m}\Omega \quad (16)$$

Converting $|Z_k|$ to Ohms (0.06582 Ω) and applying the IEC 60909 equivalent voltage source equation:

$$I''_{k3} = \frac{c \cdot U_n}{\sqrt{3} \cdot |Z_k|} \quad (17)$$

$$I''_{k3} = \frac{1.05 \cdot 400}{\sqrt{3} \cdot 0.06582} = \frac{420}{0.1140} \approx 3684 \text{ A} \quad (3.68 \text{ kA}) \quad (18)$$

This manual calculation (3.68 kA) closely benchmarks the software-simulated value (3.71 kA), with the negligible variance attributed to proprietary manufacturer reactance tables utilized by the simulation engine.

3.4 Step 4: Minimum Single-Phase Earth Fault (I''_{k1}) at the DB

To verify shock safety parameters, the minimum single-phase earth fault is calculated at the DB busbar. Per the Phase 1 methodology, minimum short-circuit calculations mandate elevated conductor temperatures to account for maximum operational resistance. A standard thermal multiplier of 1.25 is applied to the previously calculated 20°C cable resistances.

For the TN-S earthing system, the zero-sequence impedance (Z_0) is dictated by the resistance of the Protective Earth (PE) conductors.

1. Positive-Sequence Resistance (R_1)

The elevated temperature resistance of the phase conductors:

$$R_{cables_hot} = 1.25 \cdot (R_1 + R_2) = 1.25 \cdot (6.63 + 55.50) = 77.66 \text{ m}\Omega \quad (19)$$

$$R_1 = R_{source} + R_{cables_hot} = 0.965 + 77.66 = 78.63 \text{ m}\Omega \quad (20)$$

2. Zero-Sequence Resistance (R_0)

The elevated temperature resistance of the PE conductors (70 mm² and 10 mm²):

$$R_{pe_hot} = 1.25 \cdot \left(\frac{18.5 \cdot 43}{70} + 55.50 \right) = 1.25 \cdot (11.36 + 55.50) = 83.57 \text{ m}\Omega \quad (21)$$

$$R_0 = R_{pe_source} + R_{pe_hot} = 0.965 + 83.57 = 84.54 \text{ m}\Omega \quad (22)$$

3. Total Fault Loop Impedance ($|Z_{loop}|$)

Assuming zero-sequence reactance approximates positive-sequence reactance in bundled cables ($X_0 \approx X_1$), the total loop reactance is $X_{total} = X_1 + X_0 = 15.49 + 15.49 = 30.98 \text{ m}\Omega$. The total loop resistance is $R_{loop} = R_1 + R_0 = 78.63 + 84.54 = 163.17 \text{ m}\Omega$.

$$|Z_{loop}| = \sqrt{R_{loop}^2 + X_{total}^2} = \sqrt{163.17^2 + 30.98^2} \approx 166.08 \text{ m}\Omega \quad (0.1661 \Omega) \quad (23)$$

4. Calculating I''_{k1}

Applying the simplified earth fault formula with $c = 0.95$ and $U_0 = 230 \text{ V}$:

$$I''_{k1} = \frac{0.95 \cdot U_0}{|Z_{loop}|} = \frac{0.95 \cdot 230}{0.1661} = \frac{218.5}{0.1661} \approx 1315.5 \text{ A} \quad (1.32 \text{ kA}) \quad (24)$$

This mathematically validates the software's minimum earth fault output of 1.32 kA at breaker QA 51.

3.5 Step 5: End-of-Circuit Safety Verification (Hop 3)

To guarantee tenant safety, the minimum single-phase earth fault is calculated at the furthest load point. This requires adding the impedance of the final sub-circuit (Hop 3: 15 m, 4 mm² Cu for both Phase and PE) to the previously established DB busbar totals.

1. Hop 3 Conductor Resistance (Elevated Temperature)

Applying the thermal multiplier of 1.25 to the 4 mm² copper baseline:

$$R_3 = 1.25 \cdot \left(\frac{18.5 \cdot 15}{4} \right) = 1.25 \cdot 69.38 \approx 86.72 \text{ m}\Omega \quad (25)$$

Since the Phase and PE conductors share the same cross-sectional area, 86.72 mΩ is added to both the positive-sequence (R_1) and zero-sequence (R_0) totals.

2. New Total Loop Resistance (R_{loop})

$$R_1(\text{Total Phase}) = 78.63 + 86.72 = 165.35 \text{ m}\Omega \quad (26)$$

$$R_0(\text{Total PE}) = 84.54 + 86.72 = 171.26 \text{ m}\Omega \quad (27)$$

$$R_{loop} = 165.35 + 171.26 = 336.61 \text{ m}\Omega \quad (28)$$

3. New Total Loop Reactance (X_{total})

The standard reactance ($0.08 \text{ m}\Omega/\text{m}$) is applied to the 15 m Phase and PE conductors and added to the previous DB reactance total ($30.98 \text{ m}\Omega$):

$$X_3(\text{Hop 3 Phase} + \text{PE}) = (0.08 \cdot 15) \cdot 2 = 1.20 \cdot 2 = 2.40 \text{ m}\Omega \quad (29)$$

$$X_{total} = 30.98 + 2.40 = 33.38 \text{ m}\Omega \quad (30)$$

4. Total Load Fault Loop Impedance ($|Z_{loop}|$)

$$|Z_{loop}| = \sqrt{R_{loop}^2 + X_{total}^2} = \sqrt{336.61^2 + 33.38^2} \approx 338.26 \text{ m}\Omega \quad (0.338 \Omega) \quad (31)$$

5. Minimum Earth Fault Current at the Load (I''_{k1})

$$I''_{k1} = \frac{0.95 \cdot U_0}{|Z_{loop}|} = \frac{0.95 \cdot 230}{0.338} \approx 646 \text{ A} \quad (32)$$

Safety Conclusion: The specified final circuit protection is a 20 A Type C Miniature Circuit Breaker (MCB), which requires a minimum magnetic trip threshold of $10 \times I_n$ (200 A). Because the calculated minimum earth fault at the absolute end of the line (646 A) greatly exceeds 200 A, instantaneous magnetic tripping is guaranteed, ensuring optimal shock protection for the user.

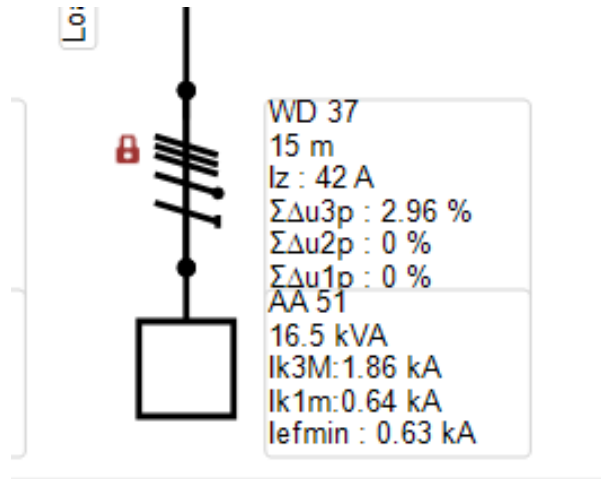


Figure 2: Ecodial simulation output at the final load (Node AA 51). The simulated minimum earth fault ($I_{k1m} = 0.64 \text{ kA}$) highly correlates with the manual calculation (646 A), verifying the end-of-circuit safety parameters.

4 Summary and Conclusion

The manual benchmark calculations executed via standard circuit analysis confirm the accuracy of the Ecodial software simulation parameters for the critical path.

Table 1 provides a direct numerical comparison between the manual verification and the software outputs at both the Distribution Board busbar and the final end-of-circuit load.

Fault Location & Type	Manual Calculation	Ecodial Simulation	Variance
DB Busbar: 3-Phase (I''_{k3})	3.68 kA	3.71 kA	-0.81%
DB Busbar: 1-Phase Earth (I''_{k1})	1.32 kA	1.32 kA	0.00%
Final Load: 1-Phase Earth (I''_{k1})	0.646 kA	0.64 kA	+0.93%

Table 1: Comparison of manual calculations vs. software simulation results.

Conclusion Statement:

The mathematical variance between the manual methodology and the software simulation is negligible (under 1% across all parameters). The minor -0.03 kA deviation in the three-phase calculation is attributed to the software utilizing proprietary, manufacturer-specific cable reactance libraries, whereas the manual calculation utilized the conservative IEC universal baseline of 0.08 m Ω /m.

Furthermore, the end-of-circuit single-phase earth fault calculation (646 A) mathematically verifies that the selected 20 A Type C final protection devices (magnetic trip threshold of 200 A) will operate instantaneously under worst-case minimum fault conditions. The calculated fault parameters ensure the protection scheme is fully compliant with IEC standard shock safety mandates.